

CS 663: Digital Image Processing

(Instructor: Suyash P. Awate)

Mid-Semester Exam: Items NOT Permitted – Books / Sheets / Electronic Devices

Roll Number: _____

Name: _____

When in doubt, make suitable assumptions, state them clearly, and proceed to solve the problem.
All the best!

1. [25 points]

Consider the task of finding parking signs in images of street scenes using “template matching”.

The only information we have about the parking sign is that always has a red ring of a known thickness and known radii (of the underlying inner circle and the outer circle). The region in the interior and the exterior of the ring can have varying patterns that are unknown (within this particular task).

The images of the street are taken by a pinhole camera that gives RGB images.

Assume that the camera is mounted on top of a car that is moving slowly along a straight road.

Assume that the direction of the road is exactly north, and the movement of the car is also exactly north.

Assume that parking signs are on boards that are always facing exactly east or exactly west.

Assume that the camera is always pointing exactly east or exactly west.

Assume that the altitude of the pinhole (in the camera) exactly equals the altitude of the parking-sign-ring's center.

Assuming that you have lots of computational and memory resources (i.e., no related constraints), design an algorithm for detecting parking signs in street images using template matching.

- (16 points) State a step-by-step high-level algorithm (pseudo-code level detail isn't needed; higher level is fine) for performing the aforementioned parking-sign detection task given a video taken from the camera of the slow-moving car.

2 marks: use of normalized cross correlation

2 marks: use of RGB information in template image and scene image.

4 marks: use of normalized cross correlation restricted within ring (ignoring regions inside and outside of ring).

4 marks: search using multiple created templates depicting red-ring ellipses (arising from perspective transformations) of varying eccentricities, but where the major axis of the ellipse is always exactly vertical in the image.

4 marks: search using multiple created templates depicting aforementioned red-ring ellipses (arising from varying distance between camera and parking sign) of varying sizes.

- (9 points) Given a specific template image and a specific scene image, give detailed implementable pseudo-code for computing the template matching for our task.
-

Please see the lecture slides, and the assignment.

2. [25 points]

Consider a convolution operation between two 1D images.

- (6 points: 3 + 3) Prove that the convolution operation is commutative. Prove that the convolution operation is distributive.

Please see the lecture slides.

-
- (7 points) Prove that the convolution operation is associative.

Please see the lecture slides.

-
- (12 points)

Consider the function $h(t) := \exp(-2t)$ when $t > 0$ (t is real valued), and $h(t) := 0$ when $t \leq 0$.

Consider another function $f(t)$ that is the box function that takes values $f(t) := 0$ when $t < 0$, $f(t) := 1$ when $0 \leq t \leq 1$, and $f(t) := 0$ when $t > 1$.

Derive the mathematical analytical expression for the convolution of $f(\cdot)$ and $h(\cdot)$, comprising only polynomial or exponential terms, devoid of any integral expressions.

Plot (show graphically) the functions $f(\cdot)$ and $h(\cdot)$, and plot the function resulting from their convolution. The plots must indicate key values on the abscissa/horizontal axis and the ordinate/vertical axis.

<https://lpsa.swarthmore.edu/Convolution/Convolution2.html#Mechanics>

3. [15 points]

- (8 points) Consider cubic-spline modeling of a function $f(x)$ over the interval $[0, 2]$ using the following data:

$$f(0) := 1, f(1) := 3, f(2) := 2.$$

Derive the system of equations underlying cubic-spline interpolation over the interval $[0, 2]$.

Assume that the curvature of the interpolating function at the interval boundaries is zero.

<https://pythonnumericalmethods.studentorg.berkeley.edu/notebooks/chapter17.03-Cubic-Spline-Interpolation.html>

- (7 points) Suppose you want to perform bilinear interpolation within an image of hue values, i.e., in an image corresponding to hue channel of the HSV representation format.

Consider that the hue values are provided at all combinations of integer abscissas and integer ordinates in a 2D plane.

Provide detailed implementable pseudo-code to compute the interpolated hue value at location $(0.3, 0.7)$.

(3 points) for converting hue values in $[0, 1]$, at each grid point, into circular coordinates (x, y) .

(3 points) for performing bilinear interpolation on circular coordinates using 3 linear interpolations, or using the direct formula.

(1 point) for converting resulting circular coordinate back to hue

4. [20 points]

- (8 points: 2 + 1 + 5) Mathematically define the structure tensor for 2D grayscale images. Prove that this structure tensor is symmetric and non-negative definite.

Please see the lecture slides.

- (5 points) Provide detailed implementable pseudo-code to compute the Harris-Stephens cornerness measure at every pixel in a 2D grayscale image.

Please see the lecture slides.

- (2 points) State and justify the mathematical condition when the aforementioned Harris-Stephens cornerness measure indicates a corner. State and justify the mathematical condition when the Harris-Stephens cornerness measure indicates an edge.

Please see the lecture slides.

- (5 points) Extend the definition of the structure tensor for 3D binary images representing segmentations of 3D objects. Clearly state the conditions for the structure tensor indicating a point that is at a corner.

gradient becomes a 3-vector.

structure tensor defined akin to that in 2D,
as a weighted average of gradient * gradient-transpose.

for a corner, all 3 eigenvalues must be significantly larger than zero

5. [20 points]

- (10 points)

Consider grayscale images A and B, each having intensities quantized to 10 distinct values, from 0 to 9.

Consider the histogram of a grayscale image A, with 10 bins, with histogram values: [0.02, 0.05, 0.08, 0.10, 0.15, 0.20, 0.18, 0.12, 0.07, 0.03].

Consider the histogram of a grayscale image B, with 10 bins, with histogram values: [0.10, 0.12, 0.08, 0.05, 0.10, 0.15, 0.20, 0.12, 0.06, 0.02].

State the algorithm to map/transform the intensities in image A in order to “match” its histogram to that of image B.

State the mapping/transformation for each intensity in image A.

Please see the lecture slides, and the assignment.

CDF A: [0.02, 0.07, 0.15, 0.25, 0.40, 0.60, 0.78, 0.90, 0.97, 1.00]

CDF B: [0.10, 0.22, 0.30, 0.35, 0.45, 0.60, 0.80, 0.92, 0.98, 1.00]

mapping: $0 \rightarrow 0, 1 \rightarrow 0, 2 \rightarrow 1, 3 \rightarrow 2, 4 \rightarrow 4, 5 \rightarrow 5, 6 \rightarrow 6, 7 \rightarrow 7, 8 \rightarrow 8, 9 \rightarrow 9$

note: no input mapped to target bin 3

- (10 points)

If we want to perform histogram equalization on image A, then state the mapping/transformation for each intensity in image A.

Please see the lecture slides, and the assignment.

Mapping:

$0 \rightarrow 0, 1 \rightarrow 0, 2 \rightarrow 1, 3 \rightarrow 2, 4 \rightarrow 3, 5 \rightarrow 5, 6 \rightarrow 7, 7 \rightarrow 8, 8 \rightarrow 9, 9 \rightarrow 9$

6. [25 points]

- (10 points)

State a method (strictly based on the material covered in class lectures) for obtaining a smooth (i.e., infinitely differentiable) estimate of the probability density function (PDF) of image intensities where the intensities are real-valued (non-quantized) scalars.

Kernel density estimation with Gaussian kernel (to ensure infinite differentiability).

Some arbitrary kernel may not work; you will lose points in that case.

See lecture slides for details.

- (15 points)

Design an algorithm that generalizes “histogram matching” to perform “PDF matching” using the smooth-PDF estimates of intensities in image A (input) and image B (reference). Consider the intensities in image A are $\{x_n^A\}_{n=1}^N$. Consider the intensities in image B are $\{x_n^B\}_{n=1}^N$.

Give an implementable pseudo-code of the algorithm.

5 points for finding CDF estimate for each intensity x_n^A in image A, using Gaussian kernel.

Assume definite integrals under Gaussian are known through library, e.g., using error functions.

5 points for formulating the optimization problem for finding the quantile in CDF of image B that matches the CDF estimate of each x_n^A in image A.

5 points for the algorithm for solving the optimization problem, e.g., using bisection method, or inversion using a high-resolution table of CDF estimates across a range of intensities in image B.
